

No. of Functional Features Included in the Solution

Date	05 July 2026	Team ID	
Project Name	Voice-Based-Concept-Understanding-Analyser	Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Voice Input & Capture	Captures live audio response input from users smoothly via web microphone.	Frontend / UI	Done	
2	Speech-to-Text Pipeline	Converts captured audio signals into formatted text transcripts utilizing speech NLP.	Backend / API	Done	
3	Concept Extraction Engine	Parses transcripts to extract key technical concepts and definitions accurately.	Backend / ML	Done	
4	Knowledge Matcher Assessment	Compares user concepts against standard reference curriculum metrics.	Backend / Core Logic	Done	
5	Analytics Dashboard	Visualizes conceptual clarity metrics, gaps, and marks contribution graphs.	Frontend / UI	Done	
6	Feedback Generator Module	Generates structured recommendations and text notes on missed keywords.	Backend / Logic	Done	
7	Error & Audio Handler	Manages low confidence speech transcripts and missing audio device exceptions.	Backend / Utils	Done	
8	User Authentication System	Secures profile storage and manages session histories across analysis requests.	Security / DB	Done	

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Audio Capture Widget, Dashboard Charts Display
2	Backend / Logic	3	Speech-to-Text Pipeline, Assessment Core Matcher, Feedback Builder
3	Database / Storage	1	User Analytics History & Profile Data tables
4	API / Integration	1	Speech Recognition API, NLP Core Concept Extraction service
5	Security / Authentication	1	Secure User Login Session Management